SOFTWARE TEST PLAN & TEST CASES

For Collaborative Study Material & Quiz Platform (CSMQP)

Version: 1.0
Governing Standard: IEEE Std 829-2008

1. Test Plan Identifier

Identifier: TP-CSMQP-V1.0-2026

2. Introduction

This test plan defines the testing scope, methodology, resources, pass/fail thresholds, and individual test cases for the Collaborative Study Material & Quiz Platform (CSMQP) web application. The primary objective is to verify that all functional workflows (materials, questions, quizzes, analytics) and non-functional guarantees conform strictly to the System Requirements Specification (SRS).

3. Test Items

- **Authentication Module (v1.0):** User registration, login, and JWT validation.
- **Material Management Engine (v1.0):** Multipart file uploader, size checker, and search filters.
- **Question Bank Engine (v1.0):** Question insertion, schema validation, and retrieval.
- **WebSocket Quiz Coordinator (v1.0):** Room lobby logic, timer broadcasts, and score calculators.
- **Performance Analytics Aggregator (v1.0):** Historical log calculations and metric delivery.

4. Features to be Tested / Not Tested

4.1 Features to be Tested

- Positive and boundary behavior of user login and access control.
- Enforcing file size boundaries on study material uploads.
- Tag and keyword search filtering for study notes and questions.
- WebSocket state fan-out latency and score computation integrity.
- Accuracy calculations and cohort percentile reporting in the analytics module.
- Pre-submission quiz answer suppression and database injection prevention.

4.2 Features Not to be Tested

- Physical network hardware failure recovery at hosting facilities.
- Performance on unlisted legacy web browsers (e.g., Internet Explorer 11).

5. Test Approach & Methodology

- **Unit Testing:** Framework-level validation of business algorithms, Bcrypt hash validation, and score calculation functions using Jest / PyTest.
- **Integration Testing:** API contract validation using Supertest / Postman collections verifying HTTP status codes and database updates.
- **Real-Time & Load Testing:** Multi-client concurrency and WebSocket latency benchmarking using Locust and k6.
- **Suspension / Resumption Criteria:** Testing will halt if core authentication fails completely (blocking dependent functional tests) and resume once an authenticating build passes smoke checks.

5.1 Security Validation Strategy

- **Token Tampering / IDOR:** Sending forged JWT payloads and altering user IDs in URL parameters to verify that foreign performance records cannot be accessed.
- **Data Leakage in Transit:** Intercepting WebSocket packets using browser developer tools during active quizzes to ensure answer keys are absent prior to round completion.
- **Injection Attacks:** Injecting SQL payloads (`' OR 1=1; --`) into search fields and question input boxes to confirm parameter binding defenses.

6. Detailed Test Cases (IEEE 829 Standard Table)

Test Case ID	Traceable Req	Test Scenario / Objective	Input / Execution Steps	Expected Result	Test Type
TC-01	FR-01	Valid User Authentication	POST `/api/v1/auth/login` with valid email and password	200 OK; valid signed JWT returned; user profile retrieved	Functional (Positive)
TC-02	FR-01	Invalid Login Rejection	POST `/api/v1/auth/login` with incorrect password	401 Unauthorized; token withheld; error message returned	Functional (Negative)
TC-03	FR-02	Valid Material Upload	Upload 8.4 MB PDF with course tag 'CS301' and title 'Virtual Memory'	201 Created; metadata stored in DB; file available in storage	Functional
TC-04	FR-02	Oversized File Rejection	Attempt upload of 28 MB PDF	413 Payload Too Large; upload aborted; storage unaffected	Functional (Boundary)

Test Case ID	Traceable Req	Test Scenario / Objective	Input / Execution Steps	Expected Result	Test Type
TC-05	FR-03	Question Bank Filter Query	GET `/api/v1/questions/search?topic=OS&course=CS301`	200 OK; returns only matching question records	Functional (Query)
TC-06	FR-04, NFR-02	Live Quiz Sync Latency	Host client emits `START_QUIZ` to 3 connected peers	All clients receive Q1 payload simultaneously within <= 150 ms	Real-Time Integration
TC-07	FR-05	Score & Accuracy Calculation	Submit quiz attempt with 7 correct and 3 wrong answers	Database records attempt accuracy as 70%	Functional
TC-08	FR-05, NFR-01	Analytics Query Latency	GET `/api/v1/analytics/performance/me` on user with 1,000 attempts	200 OK; analytics breakdown rendered in <= 600 ms	Non-Functional (Load)
TC-09	SEC-01, 5.1	Analytics IDOR Validation	User A attempts GET `/api/v1/analytics/performance/UserB_ID` with User A's JWT token	403 Forbidden; cross-user analytics access blocked	Security
TC-10	SEC-02, 5.1	In-Flight Answer Concealment	Inspect WebSocket frames during active `QUESTION_OPEN` state	Answer key and explanation completely omitted from payload	Security
TC-11	NFR-03	Platform Uptime & Availability Soak Test	Run 24-hour continuous automated health check probe (`GET /healthz`) at 30-second intervals under simulated background synthetic user traffic (50 concurrent virtual users via Locust/k6).	System maintains >= 99.5% successful response rate without unhandled daemon crashes or unrecoverable memory leaks; DB WAL logs commit cleanly.	Non-Functional (Reliability / Soak)
TC-12	SEC-01, 5.1	SQL Injection Attack Defense on Search & Input Fields	Submit malicious SQL payloads (e.g., `' OR '1'='1'; --`, `'; DROP TABLE users; --`) into `GET /api/v1/questions/search?topic=` query parameter and `prompt` field in `POST /api/v1/questions`.	Backend sanitizes inputs via parameterized SQL statements; requests either return standard filtered 200 OK with no unauthorized records exposed or 400 Bad Request; database schema and records remain intact.	Security